

Advanced Resource & Clock (ARC) Estimator

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Overview: What this script is all about ...

- This is a Synplify® Premier feature that supports Xilinx Virtex-5 and later devices that includes the 7 Series devices.
- The ARC estimator script is a Tcl based approach to do ultra-fast, early-prediction of resources & clocks for Xilinx devices based on querying the srs database.
- The biggest advantages of this script are
 - Extremely easy to use and understand reports.
 - Extremely fast to run.
 - The ability to analyze and make decisions very early in the flow.
- The current version of the script does
 - Resource usage/estimation and attribute generation for blockrams, dsp, and shift-registers.
 - Estimates all internal and primary clock nets with fanout information and generates attributes for control over BUFG insertion.

Current Features: What the script can do ...

- Estimates BRAM utilization for the design. Full details are dumped to a csv file that can be opened/manipulated using MS-Excel.
- Estimates DSP48 utilization for the design. Full details are dumped to a csv file that can be opened/manipulated using MS-Excel.
- Estimates SRL32/16 utilization for the design. Full details are dumped to a csv file that can be opened/manipulated using MS-Excel.
- Estimates all internal and primary clocks with clock driver and fanout information. Full details are dumped to a csv file.
- Generates fdc file with block_ram attributes by specifying a threshold for the blockram size.
- Generates fdc file with DSP48 attributes by specifying a threshold for the multiplier output size.
- Generates fdc with attributes for BUFG insertion and controlling BUFG inference.
- BRAM and DSP utilization can be run in two modes – namely default and attribute mode.
 - Default mode estimates default inference by mapper, i.e. without attributes.
 - Attribute mode estimates inference with global syn_ramstyle=block_ram or syn_dspstyle=dsp48 attributes.
- A summary file is created with brief information and pointer to detailed report.
 - **resource_summay.txt** for BRAM, DSP, SRL estimation
 - **clock_summary.txt** for clock estimation

Commands Usage: How to use the script ...

- Step 1 : Source the script from the Synplify Premier Tcl window
 - `source_encrypted_tcl <path/arc_est.tcl>`
- Step 2 : Open srs view
- Step 3 : Run commands from Tcl window with active srs view
 - `report_usage bram def` for bram estimation in default mode
 - `report_usage bram atr` for bram estimation in attribute mode
 - Creates report.csv with details of bram inference.
 - Outputs messages to console regarding utilization details.
 - `report_usage dsp def` for DSP estimation in default mode
 - `report_usage dsp atr` for DSP estimation in attribute mode
 - Creates report.csv with details of dsp inference.
 - Outputs messages to console regarding utilization details.
 - `report_usage srl def` for SRL estimation in default mode
 - Creates report.csv with details of srl inference.
 - Outputs messages to console regarding utilization details.
 - `report_usage all def`
 - Creates report for bram, dsp, and srl in default mode within the same report file.

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- **fdc_gen bram <threshold>**
 - Creates ramstyle.fdc with block_ram attributes for rams/roms above 'threshold' size.
- **fdc_gen dsp <threshold>**
 - Creates dspstyle.fdc with dsp48 attributes for multipliers/adders with output width greater than 'threshold'.
 - <threshold> is an integer value*
- **report_clocks <fanout>**
 - Generates clock estimation report for all clocks in the design and dumps it to report_clocks.csv.
 - Generates clock_buffer.fdc constraint file with BUFG inference control attributes for clocks with load greater than or equal to <threshold>.
 - <fanout> is a integer value that applies only for attribute generation and not for reporting.*
 - To disable/prevent BUFG inference on a clock, disable syn_insert_buffer and enable syn_noclockbuf in clock_buffer.fdc file.

Estimation Accuracy: How good is estimation ...

➤ **BRAM**

- Detects SP/SDP/TDP rams
- Detects ROMs
- Detects reset type of ram/rom op/addr registers
- Estimates based on both depth and width splitting

➤ **DSP48**

- Detects Signed/Unsigned multiplier and individual adders
- Detects mult-add structures
- Detects simple counter, loadable counter and counter with reset

➤ **SRL**

- Detects shift registers extracted as 'seqshift', registers with feedback, or single-bit flip-flop chains
- Detects reset type and presence of enable

➤ **Clocks**

- Detects all true clock nets, i.e. clock nets with sequential load only. Intermediate clock nets are eliminated.
- Detects primary clocks, i.e. clocks driven by top-level-ports/instantiated-IBUFG
- Detects internal clocks, i.e. derived clocks, generated clocks, gated clocks

Estimation may be wrong if logic is removed by the mapper due to optimizations.

Snapshot of resource_summary.txt and clock_summary.txt

```
1 Script started on Mon Dec 02 10:28:01 IST 2013
2
3
4 ##### BLOCKRAM UTILIZATION #####
5
6 BlockRam distribution:
7 >> SimpleDualPort/SinglePort rams    -> 188 brams
8 >> TrueDualPort rams                  -> 0 brams
9 >> ROMs                               -> 144 brams
10 >> Instatiated Brams                  -> 0
11
12 Total Estimated 'default' bram utilization : 332
13
14 ##### DSP UTILIZATION #####
15
16 DSP distribution:
17 >> Multipliers/MultAdd    -> 1536 DSPs
18 >> Counters                -> 134 DSPs
19 >> Adders                   -> 0 DSPs
20 >> Instantiated DSPs       -> 0
21 Total Estimated 'default' DSP utilization is : 1670
22
23 ##### SHIFT-REG UTILIZATION #####
24
25 SRL distribution:
26 >> Inferred SRL32          -> 388
27 >> Inferred SRL16          -> 2086
28 >> Instantiated SRLs       -> 0
29 Total Estimated SRL utilization : 2474
30
31 Detailed report report.csv available at /slowfs/sbgindia_cae
32
33 Script took 53 seconds to run
```

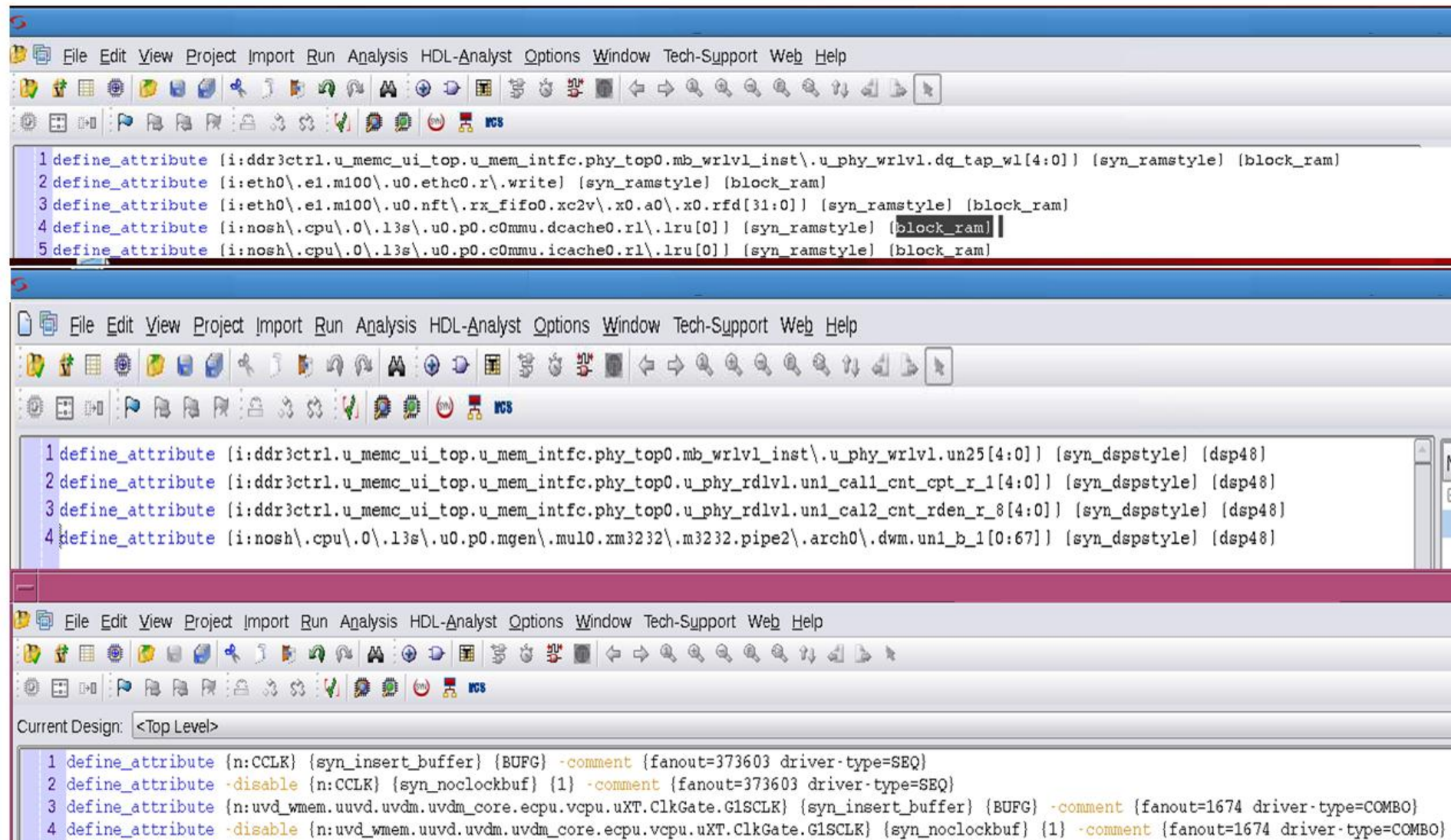
```
1 Script started on Sat Nov 23 15:26:42 IST 2013
2
3
4 ##### Clock Utilization Report #####
5
6
7 Clock fanout threshold set by user - 100
8
9 Primary Clocks:
10 >> Total no. of Clocks with PORT driver          -> 1
11 >> Total no. of Clocks with instantiated IBUFG driver -> 1
12
13 Internal Clocks:
14 >> Total no. of Clocks with sequential driver      -> 1
15 >> Total no. of Clocks with combinational driver   -> 79
16 >> Total no. of Clocks with instantiated BUFG driver -> 9
17
18 There are a total of 91 clocks in the design
19
20 22 clocks are with load >= 100
21
22 Detailed report report_clocks.csv is available at /slowfs/sbgindia_cae
23
24 Constraint file clock_buffer.fdc with clock-buffer-insertion-control a
25
26 Script took 4 minutes to run
```

Snapshot of report.csv & report_clocks.csv

	A	B	C	D	E
1	#### BLOCKRAM UTILIZATION ####				
2	TYPE	NAME	RAM-SIZE	BRAMS_NEEDED	
18	RAM	i:i_BEP.hsd_switch_lite_inst.I_ROUT.I_RAM.ram[71:0]	147456	4	
19	RAM	i:i_BEP.hsd_switch_lite_inst.I_ROUT.node_ccount[3:0]	1024	1	
20	RAM	i:i_DDR3.i_Phy.c0_u_memc_ui_top.u_mem_intfc.phy_top0.mb_wrlvl_inst\ .u_phy_wrlvl.dq_tap_wl[	20	0	
21	TDP-RAM	i:i_DDR3.i_Tester.dpram[127:0]	20	4	
22	TDP-RAM	i:i_DDR3.i_Tester.dpram_1[127:0]	20	4	
6	ROM	i:i_Video.i_Debug_SDI.i_GTX.i_Tx_Output.EDH.DEC.ad.inc_loops	16384	1	
7	ROM	i:i_Video.i_Debug_SDI.i_GTX.i_Tx_Output.EDH.EDHRX.check_parity	32	0	
90	#### DSP UTILIZATION ####				
91	TYPE	NAME	MULTIPLIER O/F	DSPs_NEEDED	
92	unsigned-mult	i:i_BEP.backplane_serial_if_inst.backplane_serial_if_regif_inst.rx_raddr[8:0]	9	1	
93	unsigned-mult	i:i_BEP.backplane_serial_if_inst.backplane_serial_if_regif_inst.rx_waddr[8:0]	9	1	
94	unsigned-mult	i:i_BEP.backplane_serial_if_inst.backplane_serial_if_regif_inst.tx_raddr[8:0]	9	1	
95	unsigned-mult	i:i_BEP.backplane_serial_if_inst.backplane_serial_if_regif_inst.tx_waddr[8:0]	9	1	
96	unsigned-mult	i:i_BEP.hsd_switch_lite_inst.I_ROUT.un1_buf_raddr[11:0]	12	1	
97	unsigned-mult	i:i_BEP.hsd_switch_lite_inst.I_ROUT.un1_buf_waddr_2[11:0]	12	1	
98	unsigned-mult	i:i_VBI.i_VBI_Chain_0.i_VBI_Extract.i_Analogue.i_AMOL.un1_amol_bit_numberx4096_plus2048[18:4	15	1	
99	unsigned-mult	i:i_VBI.i_VBI_Chain_0.i_VBI_Extract.i_Analogue.i_CC.un1_cc_bit_numberx4096_plus2048[16:2]	15	1	
100	unsigned-mult	i:i_VBI.i_VBI_Chain_0.i_VBI_Extract.i_Analogue.i_Input.un1_luma3[31:18]	14	1	
101	signed-mult	i:i_VBI.i_VBI_Chain_0.i_VBI_Extract.i_Analogue.i_Input.un1_prod7[31:19]	13	1	
102	signed-mult	i:i_VBI.i_VBI_Chain_0.i_VBI_Extract.i_Analogue.i_Input.un1_prod16[31:22]	10	1	

1	##### Clock Utilization Report #####			
2	ClkDriverType	ClkDriver	ClkNet	Fanout
3	SEQ	i:i_tipihc_isclockgeneration.I_3.I_2	n:CCLK	373603
4	COMBO	i:uud_wmem.uuud.uudm.uudm_core.ecpu.vcp	n:uud_wmem.uuud.uudm.uudm_core.ecpu.v	1674
5	COMBO	i:uud_wmem.uuud.uudm.uudm_core.ecpu.vcp	n:uud_wmem.uuud.uudm.uudm_core.ecpu.v	124
6	IBUFG	i:N_1	n:N_1_c	1
7	BUFG	i:umrbus_pllinst.clkoutbufg	n:i_tipihc_isbuscontrol.I_1.N_1	1862
8	PORT	p:sys_clock	n:i_tipihc_isclockgeneration.I_1.N_1	482

Snapshots of generated FDC files



Limitations

- BRAM estimation does not detect byte-enabled RAM.
- Presence of blackboxes/encryptedIPs/encryptedDW might cause big differences in estimation.
- SRL estimation occurs for default mode only.
- Attribute generation for SRL is not supported yet. (*Use attribute `syn_srl_mindepth` for setting SRL inference threshold.*)
- Clock estimation does not include clocks generated by blackboxes or dedicated blocks like GTX/GTP, for example.
- Devices prior to Virtex-5 are not supported.